

# SUPER CODER 3.0

## INNOVATORS' SERIES



Name \_\_\_\_\_

Class \_\_\_\_\_ Section \_\_\_\_\_

Roll No. \_\_\_\_\_ Phone No \_\_\_\_\_

School \_\_\_\_\_



**SuperCoder 3.0 | First Edition 2022 | Revised Edition 2025**

As per latest guidelines given in National Skills Qualifications Framework  
and the Revised Skill Education Curriculum

**Aligned with NEP 2020, NCF 2023**

# PREFACE

In the rapidly evolving digital landscape of the 21st century, technological proficiency has become a cornerstone of modern education and professional development. As our world becomes increasingly interconnected and digitally driven, the ability to understand, navigate, and create within the digital realm is no longer a luxury, but a necessity. This shift demands a comprehensive approach to both digital literacy and coding education.

**SuperCoder 3.0**, developed by SuperTeacher Edureforms Pvt Ltd, offers such a curriculum designed to meet this growing need for digital competencies. This course is founded on robust educational principles, based on international standards (CSTA and ISTE), and aligned with the National Skills Qualifications Framework, the Revised Skill Education Curriculum, India's National Education Policy 2020, and the National Curriculum Framework 2023. It employs a progressive learning approach, with each level building upon the previous one.

At its core lies the **C4 model** - Context, Connection, Concept, and Competency - the load-bearing pillars that support not just the retention of information, but its practical application in the real world. This is not mere rote learning, but a living, breathing understanding of technology's place in our lives.

SuperCoder 3.0 encompasses a broad spectrum of digital knowledge areas: Digital Literacy for foundational skills, Programming and Computational Thinking for logical problem-solving, Artificial Intelligence for understanding intelligent systems, Data Science for extracting insights, Digital Citizenship and Cybersecurity for safe navigation, and Gaming and Game Development as a powerful dimension that harnesses interactive entertainment to teach design thinking, creative problem-solving, and advanced programming concepts through engaging, hands-on experiences.

It also includes skill modules recommended by experts, ensuring that the approach produces well-rounded digital citizens equipped to navigate the complexities of our society. The skills and knowledge imparted through SuperCoder 3.0 serve as the foundational elements for future innovations. They provide learners with the tools to conceptualize, design, and implement technological solutions to complex problems.

With **SuperCoder 3.0** as your blueprint, what wonders will you build? What problems will you solve? What future will you create? We invite you on this educational journey!

The SuperCoder Team  
SuperTeacher Edureforms Pvt. Ltd

## OUR TEAM

### Authors

Panneru Bhanu Prasad  
P Vishal Naidu  
Vivek Avaduta  
Rahul Choudhary  
Subba Rao Desai

### Research and Development Team

Vasudevan Natarajan  
Swathi Ambati  
Lukesh Sahu

### Illustrations & Images

[www.freepik.com](http://www.freepik.com)

### Creative Team

Mani Preetham Kuppala  
Budala Romel Lymon  
Chinthapally Venkatesh  
Bandampally Sunny  
Jindam Harshith Kumar

# CONTENT FEATURES

Hi! I'm AI-Ra



I am a new creation, inspired by the A.I (Artificial Intelligence) and Ra, the Egyptian god of sun and light, both of which represent 'knowledge'. I'll be with you throughout the book, explaining the concepts in a flowing, lucid way. This textbook is written using the C4 Model: Context Setting, Connecting to Real Life, Concept Exploration, and Competency Building.



The chapters begin with context-setting: a question which is answered at the end. It can be a problem statement, or a topic-related intriguing question to spark the readers' curiosity and learning goals of the chapter in brief.

InfoByte



The InfoBytes contain interesting facts related to the topic to hook the learners to the lesson.

try < it >



There will be many 'Try it's' sprinkled in between the chapter body. They are simple activities to keep the students' minds stimulated.



This definition boxes are used whenever a new term is introduced. These define the term, without breaking the flow of the lesson.

LET'S RECAP!

'Let's Recap' summarizes the important points taught in the chapter.

WORKSHEET



The worksheet has varied types of questions, to assess the learning. The questions check the competencies of the learners, rather than their rote-learning ability.

ACTIVITY JUNCTION!

'Activity Junction' directs the students to explore and improve their skills, in context of the topics taught in the chapter.

*This book series integrates NEP, SDGs, Competency-Based Education frameworks, CSTA and ISTE standards throughout its content, aligning with modern educational approaches and global standards while fostering comprehensive skill development.*

### DOMAIN I : DIGITAL LITERACY

#### 1. CLICK TO CONNECT - Basics of Internet and Networking 5

- Computer networks
- The Internet
- Network topology
- Types of computer networks
- Working of websites

### DOMAIN II : PROGRAMMING AND COMPUTATIONAL THINKING

#### BRIDGE IT - Introduction to Python 21

#### 2. FUNCTION POWER WITH TURTLE - Exploring Python Libraries 25

- Functions in Python
- [Project 2: Draw a Smiley](#)
- Introduction to Turtle
- Modules in Python
- Libraries in Python
- [Project 1: Area and Circumference](#)

#### 3. NESTED FOR A WHILE IN A LOOP - Making patterns with Turtle Library 56

- If and If-else
- While and While-else
- [Project 1: Polygon Maker](#)
- If-elif-else
- For loop and Nested loops
- [Project 2: Planets around the Sun](#)

#### 4. EVENTS WITH TURTLE - Interactive programs using Events and Turtle 82

- Events in Python
- Key and Mouse Events
- Binding and Callback Functions
- onkey(), onkeypress(), and onclick()

#### 5. STYLISH WEBSITES, MADE SIMPLE - Exploring HTML and CSS 98

- HTML tags
- CSS (Inline, Internal, and External)
- HTML attributes
- [Project: Snowy Owl](#)

### DOMAIN III : ARTIFICIAL INTELLIGENCE

#### 6. MAKING OF AI - AI Project Cycle and Ethics 124

- AI Project Cycle
- 4Ws Problem Canvas
- Stages of AI Project Cycle
- AI Ethics

### DOMAIN IV : DATA SCIENCE

#### 7. THINK LIKE A DATA SCIENTIST - Data Science Algorithms 143

- Data Analysis
- Algorithms
- Data representation

#### 8. TALKING TO DATA - Introduction to DBMS 158

- Data and Information
- DBMS (Database Management System)
- Understanding LibreOffice Base
- [Project: School Records](#)
- What is a Database?